D 122515 **(Pages : 5)** **Name.....**

Reg. No.....

SECOND SEMESTER M.Sc. DEGREE (REGULAR/SUPPLEMENTARY) EXAMINATION, APRIL 2025

(CBCSS)

Mathematics

MTH 2C 09—ODE AND CALCULUS OF VARIATIONS

(2019 Admission onwards)

Time : Three Hours Maximum : 30 Weightage

Section A (Short Answer Type Questions)

Answer **all** *questions.*

Each question carries a weightage 1.

1. Find the series solution of the differential equation $y' = xy$.

2. What it mean that $x = x_0$ is a regular singular point of the differential equation

$$y'' + \mathrm{P}(x)\, y' + \mathrm{Q}(x)\, y = 0.$$

3. Determine the nature of the point $x = 0$ for the differential equation

$$y'' + (\sin x)\, y = 0.$$

4. Prove that $\frac{d}{dx} \mathrm{J}_0(x) = -\mathrm{J}_1(x)$.

5. Find the critical points of system

$$\begin{cases} \frac{dx}{dt} = y^2 - 5x + 6 \\ \frac{dy}{dt} = x - y \end{cases}.$$

6. Is the function $x^2 - xy - y^2$ positive definite ? Justify your answer.

Turn over

7. Find the exact solution of the initital value problem

$$y' = 2x\left(1 + y\right), y\left(0\right) = 1.$$

8. State a sufficient condition ensuring the local existence and uniqueness of a solution for an initial value problem

$$y' = f\left(x, y\right), y\left(x_0\right) = y_0.$$

(8 × 1 = 8 weightage)

Part B (Paragraph Type Questions)

Answer any **two** *questions from each module.*
Each question carries a weightage 2.

MODULE I

9. Find a series solution of the differential equation $\left(1 + x^2\right) y'' + 2xy' - 2y = 0.$

10. Prove that the differential equation

$$4x^2y'' - 8x^2y' + \left(4x^2 + 1\right) = 0.$$

has only one Frobenius series solution. Also find the general solution.

11. Prove that

$$x\left[\lim_{a \to \infty} \mathrm{F}\left(a, a, \frac{3}{2}, \frac{-x^2}{4a^2}\right)\right] = \sin x.$$

MODULE II

12. Find the general solution of the system

$$\begin{cases} \dfrac{dx}{dt} = -3x + 4y \\ \dfrac{dy}{dt} = -2x + 3y \end{cases}.$$

13. Describe the phase portrait of the system

$$\begin{cases} \dfrac{dx}{dt} = 0 \\ \dfrac{dy}{dt} = 0 \end{cases}.$$

14. Show that $(0, 0)$ is an asymptotically stable critical point of

$$\begin{cases} \dfrac{dx}{dt} = -y - x^3 \\ \dfrac{dy}{dt} = x - y^3 \end{cases}$$

MODULE III

15. For what points (x_0, y_0) does the initial value problem

$$y' = y|y|, \quad y(x_0) = y_0$$

have a unique solution on some interval $|x - x_0| \leq h$.

16. Find the extremals for the integral

$$\int_{x_1}^{x_2} f(x, y, y')\, dx$$

if :

$$f(x, y, y') = y^2 - (y')^2.$$

17. Show that the zeros of the functions $a \sin x + b \cos x$ and $c \sin x + d \cos x$ are distinct and occur alternately whenever $ad - bc \neq 0$.

(6 × 2 = 12 weightage)

Turn over

Part C (Essay Type Questions)

Answer any **two** *from the following four questions (*18-21*).*
Each question carries a weightage 5.

18. Prove that

$$\int_{-1}^{1} \mathrm{P}_m(x)\,\mathrm{P}_n(d)\,dx = \begin{cases} 0 & \text{if } n \neq m \\ \dfrac{2}{2n+1} & \text{if } m = n \end{cases}.$$

19. (a) Prove that

$$\frac{d}{dx}\left[x^p\,\mathrm{J}_p(x)\right] = x^p\,\mathrm{J}_{p-1}(x).$$

(b) If $\mathrm{W}(t)$ is the Wronskian of the two solutions of the homogeneous system

$$\begin{cases} \dfrac{dx}{dt} = a_1(t)\,x + b_1(t)\,y \\ \dfrac{dy}{dt} = a_2(t)\,x + b_2(t)\,y \end{cases},$$

then prove that $\mathrm{W}(t)$ is identically zero or nowhere zero.

20. Consider the linear system

$$\begin{cases} \dfrac{dx}{dt} = a_1 x + b_1 y \\ \dfrac{dy}{dt} = a_2 x + b_2 y \end{cases}$$

with auxiliary equation

$$m^2 - (a_1 + b_2)\,m + (a_1 b_2 - a_2 b a_1) = 0.$$

If the roots m_1 and m_2 of the auxiliary equation are real and equal, then prove that the critical point $(0, 0)$ is a nod.

21. (a) Find the stationary function of

$$\int_0^4 \left[xy' - (y')^2 \right] dx$$

which is determined by the boundary conditions $y(0) = 0$ and $y(4) = 3$.

(b) Let $y(x)$ and $z(x)$ be non-trivial solutions of

$$y'' + q(x)\, y = 0$$

and

$$z'' + r(x)\, z = 0$$

where $q(x)$ and $r(x)$ are positive functions such that $q(x) > r(x)$ Prove that $y(x)$ vanishes at least once between any two successive zeros of $z(x)$.

(2 × 5 = 10 weightage)